

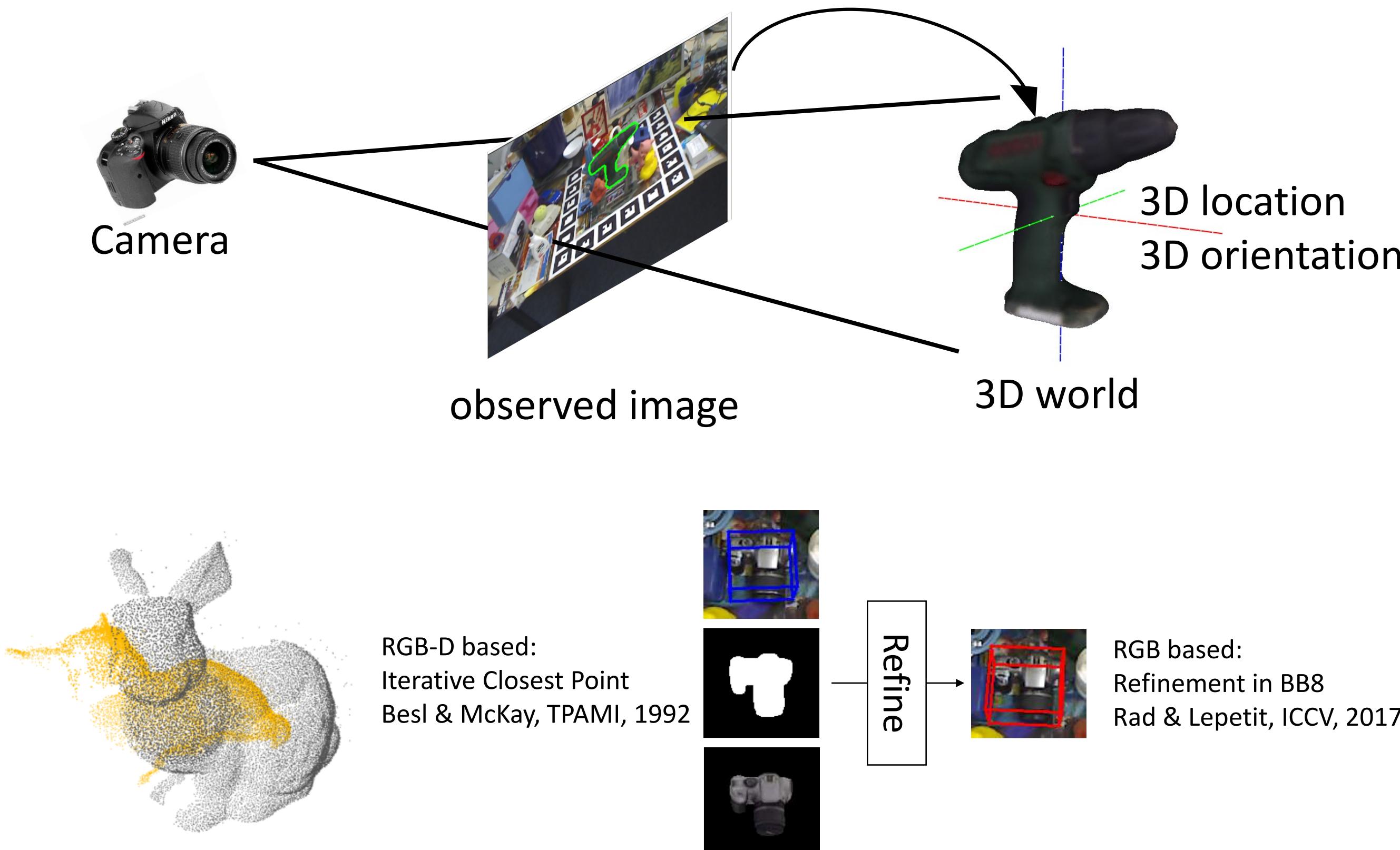
6D Object Pose Tracking with the Deep Iterative Matching Network

Lirui Wang, CSE 490 Autumn 2018

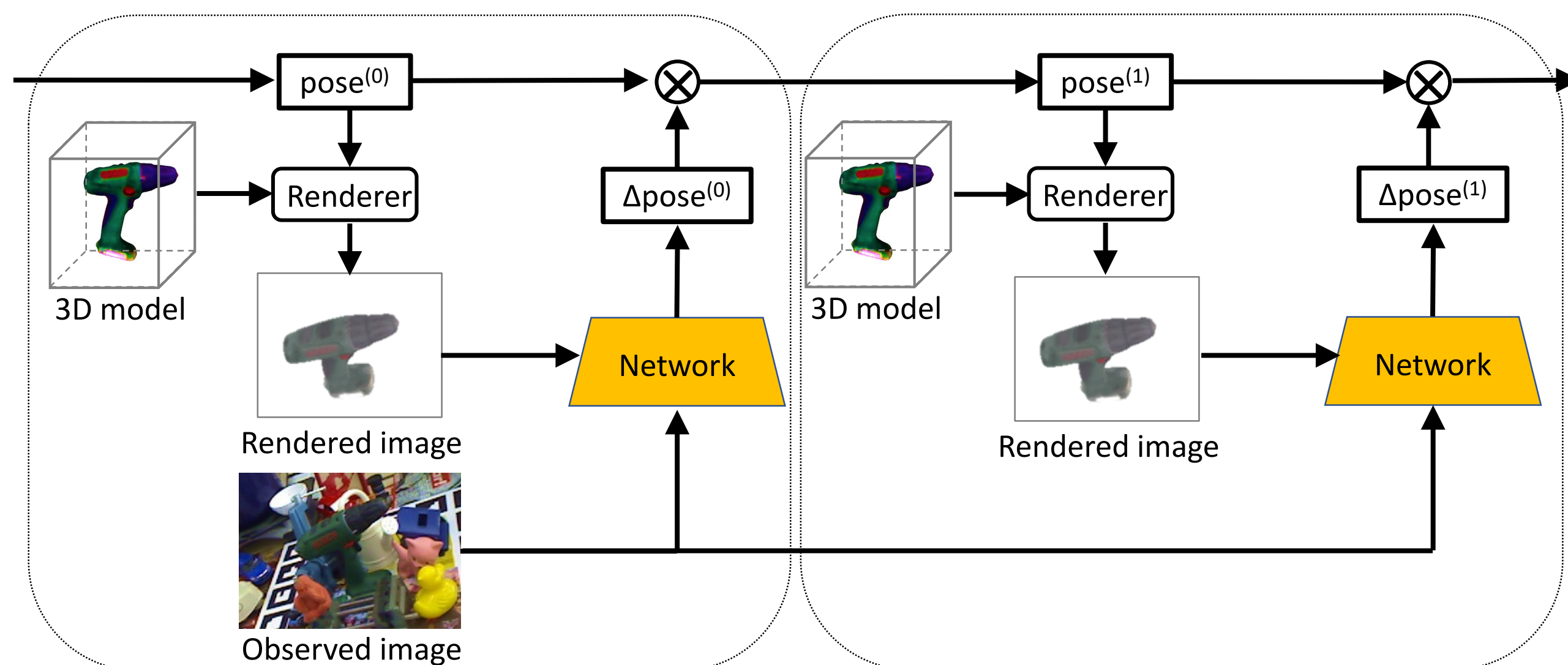
Paul G. Allen School of Computer Science & Engineering



6D Object Pose Estimation and Refinement



DeepIM: Deep Iterative Matching



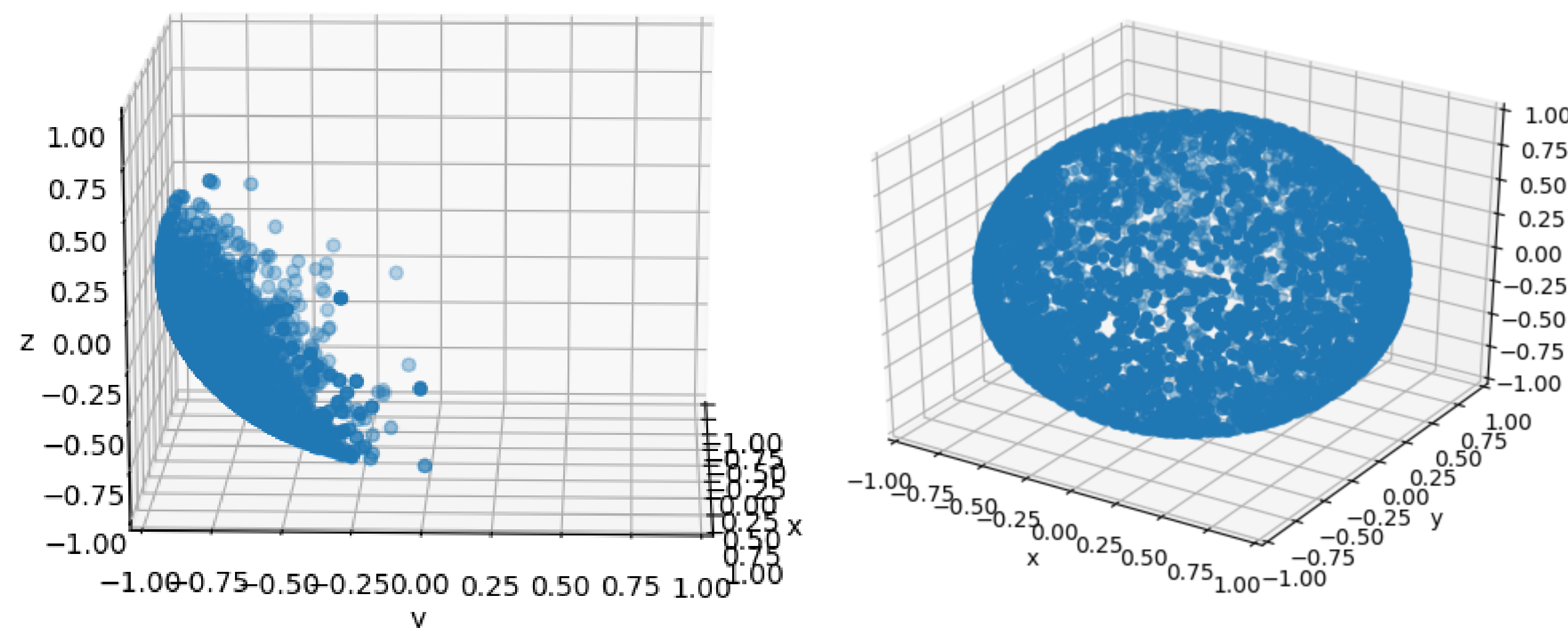
- Zoom-in Operation
- FlowNet Simple Backbone
- Untangled Transformation Representation
- Point Matching Loss

DeepIM Performance on LINEMOD

Method	Pose CNN	Pose CNN + DeepIM	FRCNN	FRCNN + DeepIM
5cm 5°	19.4	85.2	11.9	83.4
6D Pose	62.7	88.6	33.1	86.9
Proj. 2D	70.2	97.5	20.9	95.7

DeepIM: Accurate pose refinement using RGB images

Synthetic Pose Distribution



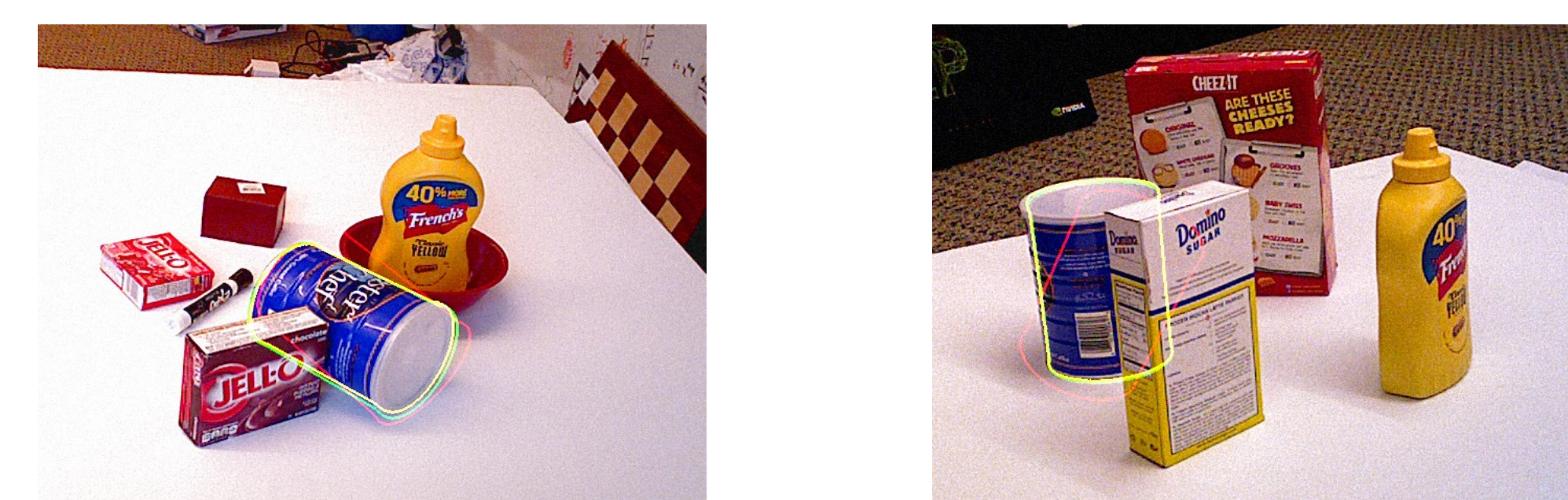
Occlusion and Synthetic Render



Algorithm For PNP Occlusion

Occlusion Algorithm (Object BB_{3d} , Occluder BB'_{3d})
 $BB_{2d} \leftarrow$ Smallest bounding box of BB_{3d} projection
 $BB'_{2d} \leftarrow$ Sample a random corner of BB_{2d} to occlude
 Match the opposite corner of BB'_{3d} to BB_{2d}
 pose of Occluder $\leftarrow$ solvePNP for BB'_{3d} and BB_{2d}

Test Set Performance



Reference

- [1] Rad et al., BB8: A scalable, accurate, robust to partial occlusion method for predicting the 3D poses of challenging objects without using depth. In ICCV, 2017.
- [2] Kehl et al., SSD-6D: Making rgb-based 3D detection and 6D pose estimation great again. In CVPR, 2017.
- [3] Y. Li, G. Wang, X. Ji, Y. Xiang, and D. Fox. Deepim: Deep iterative matching for 6d pose estimation. arXiv preprint arXiv:1804.00175, 2018.
- [4] Xiang, Y., Schmidt, T., Narayanan, V., Fox, D.: PoseCNN: A convolutional neural network for 6D object pose estimation in cluttered scenes. arXiv preprint arXiv:1711.00199.

Experiments on the YCB Dataset

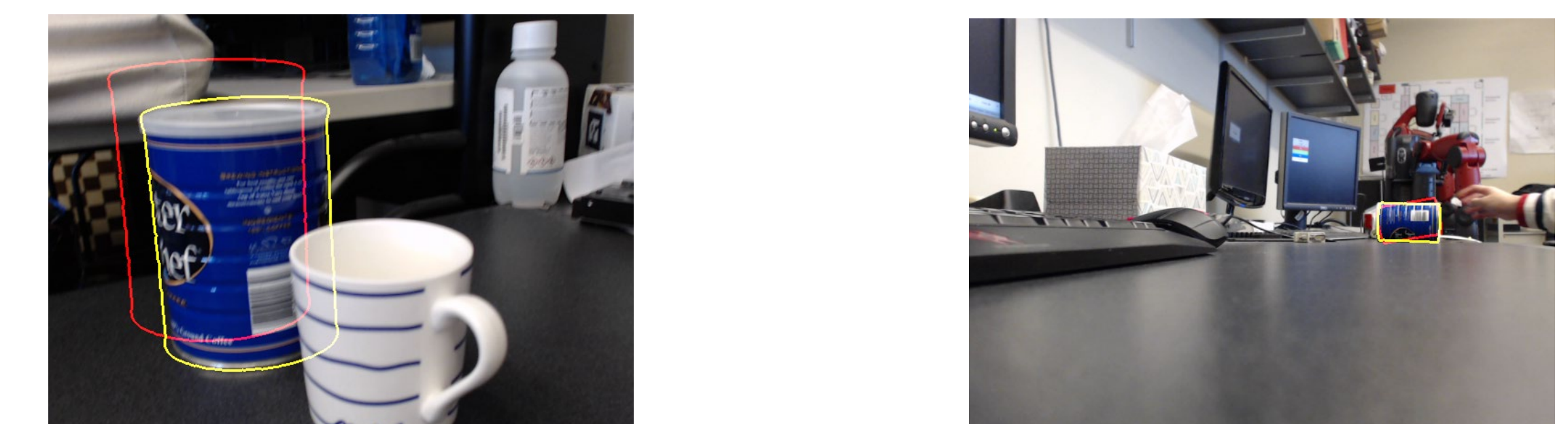
- Ablation study on pose distribution

Test Metric Method	5cm 5°	6D Pose	Proj. 2D
Perturb	0.5432	0.6391	0.4427
Random	0.8839	0.9793	0.6328

- Overall Performance (Test with 4 Iterations)

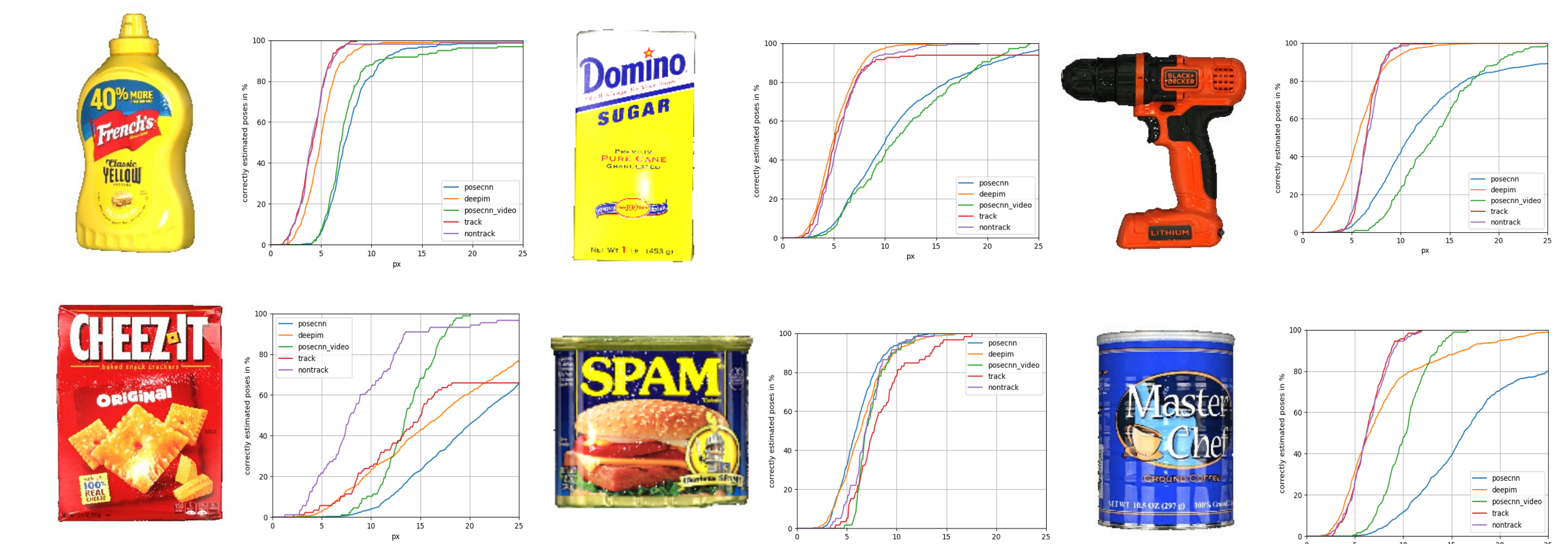
Test Metric Method	5cm 5°	6D Pose	Proj. 2D
PoseCNN	0.0374	0.1923	0.0078
Per Frame	0.8113	0.9717	0.6588
Tracking	0.7993	0.9831	0.6474
Synthetic	0.6377	0.6134	0.3723

Real Time Tracking



Real-Time Tracking Performance 10 Frames Per Second With 1 iteration refinement

Results



Future Works

- Train Well Only on Synthetic Data
- Enhance Test Speed by Faster Rendering
- On a Par with RGBD Method and ICP
- Apply on Robotic Manipulation Tasks